

Model Diagnostics Made Visual: Score Plots for Choice Modelling with Apollo

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Sebastian Büscher

Notation

$$U_{n,j} = X'_{n,j}\beta^{(X)} + Z'_{n,j}\beta_j^{(Z)} + \varepsilon_{n,j}$$

$$y_n = \arg \max_j U_{nj}$$

$$\Pr(y_n \mid \beta) = L_n(\beta)$$

$$n = 1, \dots, N$$

$$j = 1, \dots, J$$

$$ll(\beta) = \sum_{n=1}^N \log(L_n(\beta)) = \sum_{n=1}^N ll_n(\beta)$$

$$\hat{\beta}_{\text{ML}} = \arg \max_{\beta} ll(\beta)$$

$$\iff \partial_{\beta} ll(\hat{\beta}_{\text{ML}}) = \sum_{n=1}^N \partial_{\beta} ll_n(\hat{\beta}) = \sum_{n=1}^N s_n(\hat{\beta}) \stackrel{!}{=} 0$$

Gradients

Gradient information usage in **Choice Modelling**:

- **score function: estimation** of maximum likelihood estimators (score ≈ 0)
- **individual level gradients**: advanced optimization procedures like the **BGW** implemented in **apollo**¹
- **individual level gradients**: robust **sandwich estimator** for the estimator variance²
- **individual level gradients**: inform modelling of **deterministic heterogeneity**, improve **latent class structures** and guide **distribution choice** for continuous mixture models³
- **pair level gradients**: Lagrange multiplier type tests for CML estimation to test for intra-individual heterogeneity⁴

¹Bunch et al. 1993

²Qin and Lawless 1994

³Hess 2026

⁴Büscher and Bauer 2025

Gradient information usage in **Classical Linear Regression Modelling**:

- **visual model diagnosis** for **non-linear covariate dependencies** → **residual plots**
- tests for non-linear dependencies → **Ramsey RESET test**
- **visual model diagnosis** for **heteroscedasticity** → **residual plots**
- **visual model diagnosis** for **independence of the errors** → **Standardized residual plots**
- tests for independence of the errors → **Durbin-Watson test**
- **outlier detection** → **Studentized residuals** and **Cook's distance**

Residuals are just observational level gradient contributions for the intercept parameter

Lemma

Define $s_n^{(k)}(\beta; X_n, y_n) = \partial_{\beta^{(k)}} l_n(\beta)$. Let β_0 be the true parameter vector governing the DGP, and let $\hat{\beta}$ be an estimator of β_0 such that $P_{n,i}(\hat{\beta})$ is a consistent estimator of $P_{n,i}(\beta_0)$. Furthermore, assume the probabilities for all observed choice occasions n and all choice alternatives i to be bounded from below ($P_{n,i}(\beta) > \delta \forall n, i$ for some $\delta > 0$) and that the gradients of the probabilities w.r.t. β are bounded ($|\partial_{\beta^{(k)}} P_{n,i}(\beta)| < M$ for some $0 < M < \infty$). Then the score contributions conditional expected value will converge to zero as the number of observations N goes to infinity, so

$$E \left(s_n^{(k)}(\hat{\beta}; X_n, y_n) \mid X_n \right) \xrightarrow{N \rightarrow \infty} 0.$$

Proof

Sebastian Büscher (2024). “Visual Guidance for Model Specification: Introducing Score Plots for Discrete Choice Models”. In:
Available at SSRN. SSRN: 5065076

Lemma

Core Message: Average score information should be zero *let*
 $\hat{\beta}$ be an estimator **independent of the covariate values X** *Furthermore,*
assume the probabilities $P_{n,i}(\beta)$ of alternatives i to be
bounded from below ($P_{n,i}(\beta) > \delta \forall n, i$ for some $\delta > 0$) and that the gradients of the probabilities

**Idea: Visualise gradient contributions depending on covariate values
to detect model misspecifications**

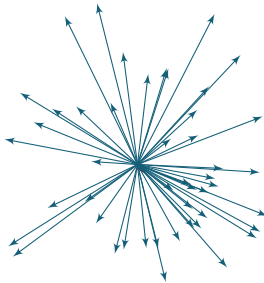
$$E \left(S_n(\beta; X_n, y_n) \mid X_n \right) \longrightarrow 0.$$

Proof

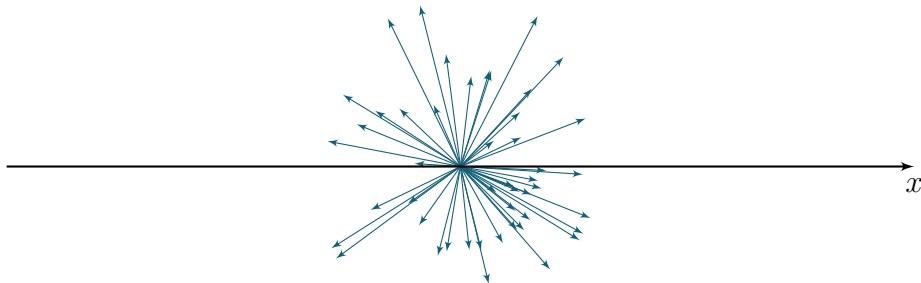
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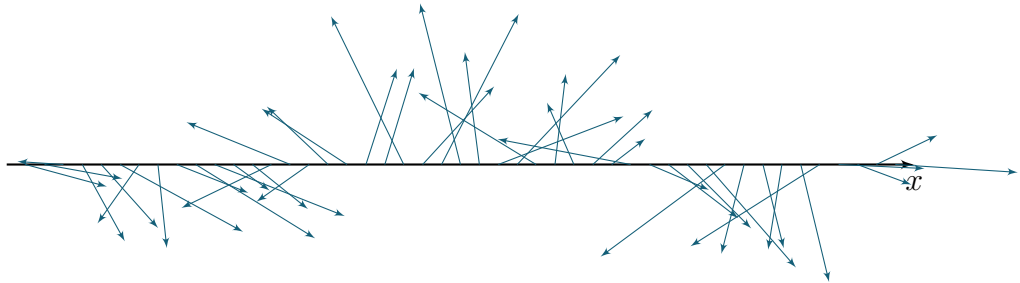
Visualisation

3. Visualisation

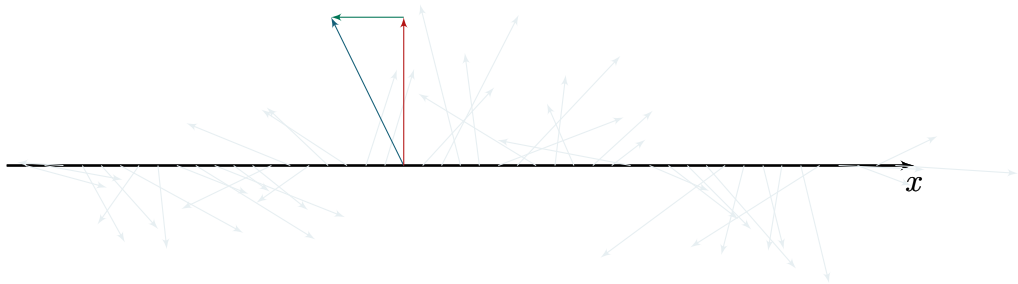


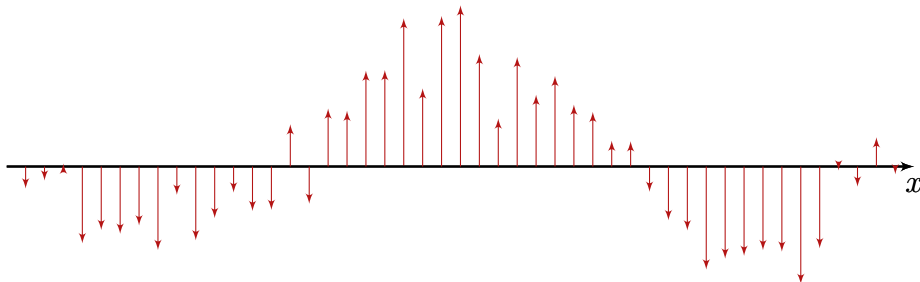
3. Visualisation





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Shiny



DGP

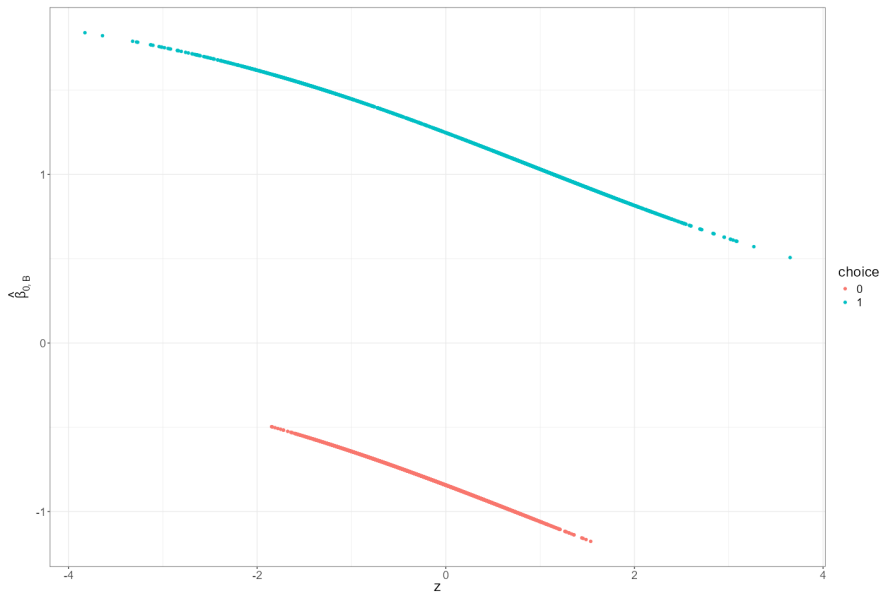
model

plot

model summary

score plot

$$\tilde{V}_I = \tilde{\beta}_{0,I} + \tilde{\beta}_{z,I} z$$

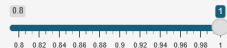


Score Plot Settings

raw gradients or score plot:

- ☒ raw gradients
- ☐ score plots

max fraction of each alternative:



min fraction of each alternative:



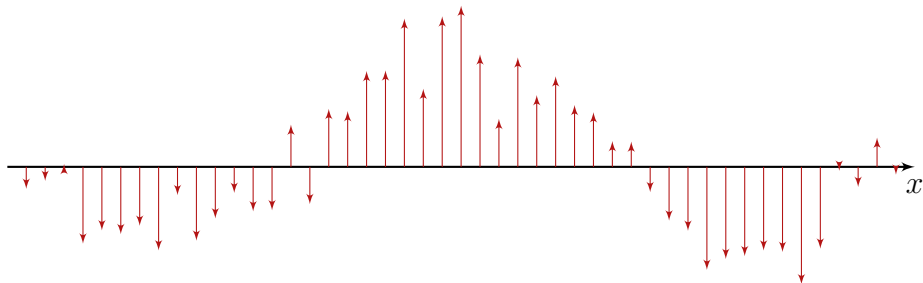
☒ add colour

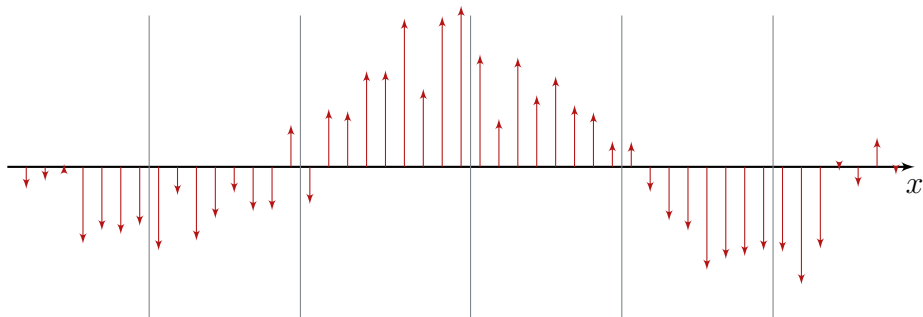
☐ add 95% CIs

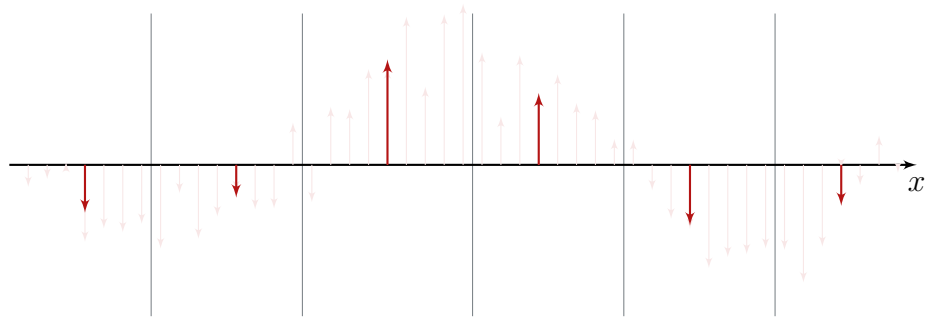
ggplot settings

☐ show DGP

Generate Score Plots









DGP

model

plot

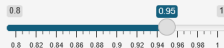
Tune Plot Behaviour

Score Plot Settings

raw gradients or score plot:

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max fraction of each alternative:



min fraction of each alternative:



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ggplot settings

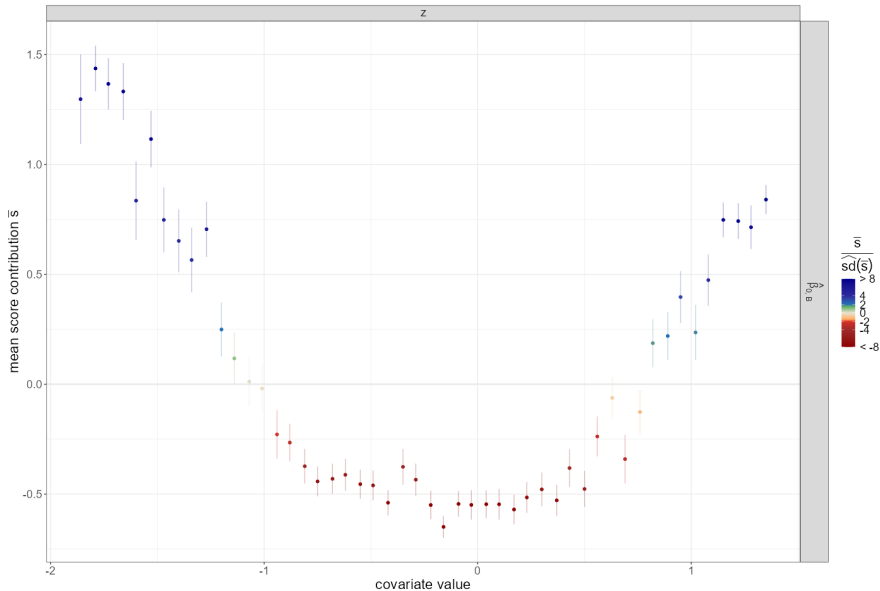
☐ show DGP

Generate Score Plots

model summary

score plot

$$\tilde{V}_i = \tilde{\beta}_{0,i} + \tilde{\beta}_{z,i} z$$





DGP

model

plot

Plot Variables

independent plot variables:

☒ x_i ☒ z

Gradient Directions $\partial_{\beta^{(\cdot)}} l l_n = s_n^{(\cdot)}$

gradient directions plotted:

☒ ASCs

☐ x_i

☐ z

Tune Plot Behaviour

[Score Plot Settings](#)

[ggplot settings](#)

☒ show DGP

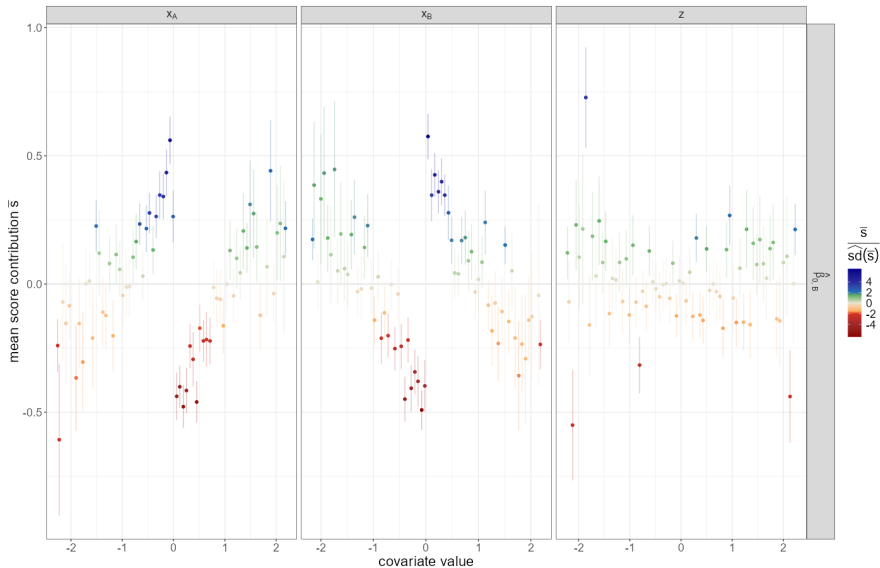
[Generate Score Plots](#)

model summary

score plot

$$\tilde{V}_i = \tilde{\beta}_{0,i} + \tilde{\beta}_x x_i + \tilde{\beta}_z z$$

$$V_i = \beta_{0,i} + \beta_x x_i + \beta_{x>0} I(x_i > 0) + \beta_z z$$





DGP

model

plot

Plot Variables

independent plot variables:

☒ x_i ☒ z

Gradient Directions $\partial_{\beta^{(\cdot)}} l_n = s_n^{(\cdot)}$

gradient directions plotted:

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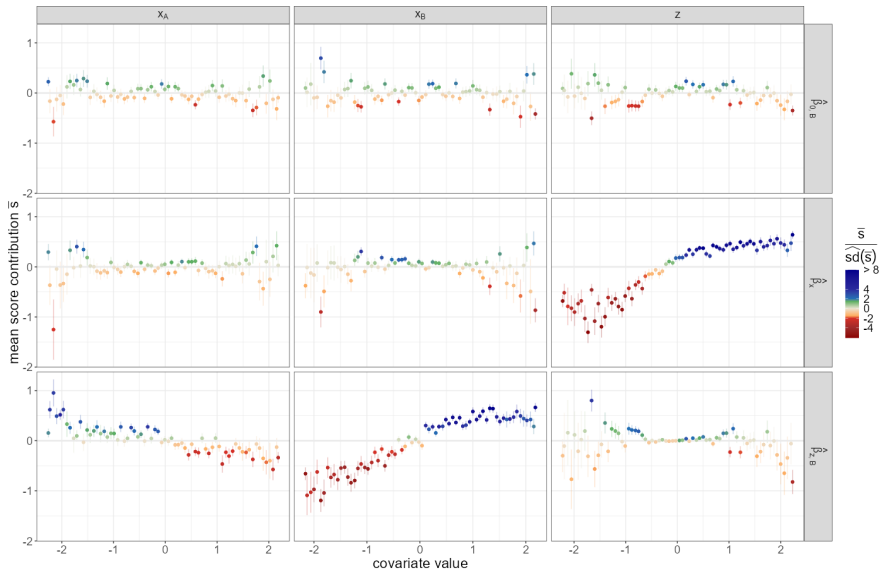
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model summary

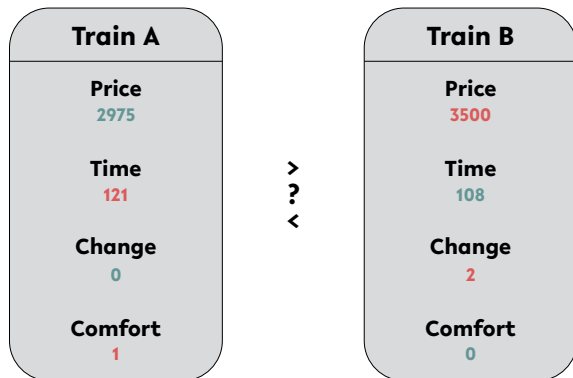
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$$\tilde{V}_i = \tilde{\beta}_{0,i} + \tilde{\beta}_x x_i + \tilde{\beta}_z z$$

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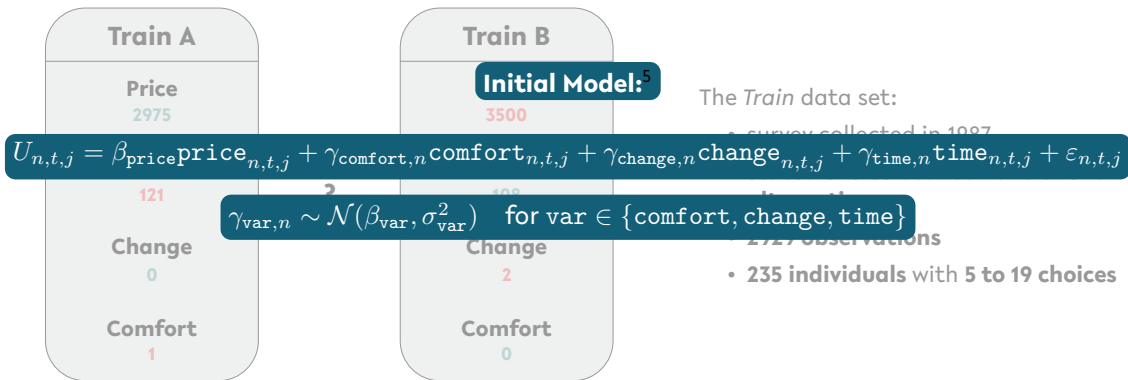


Application



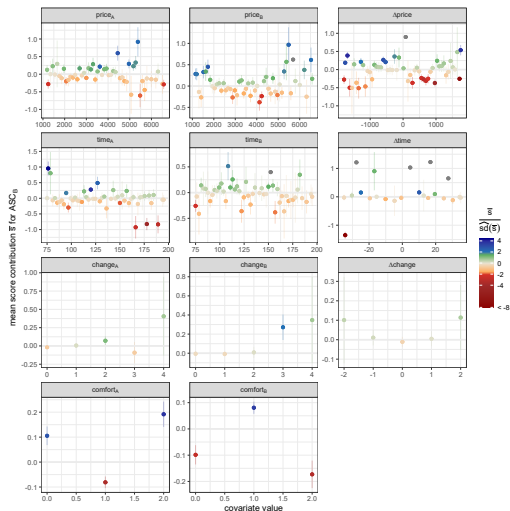
The *Train* data set:

- survey collected in 1987
- **stated choice** between two **two rail alternatives**
- **2929 observations**
- **235 individuals** with **5 to 19 choices**



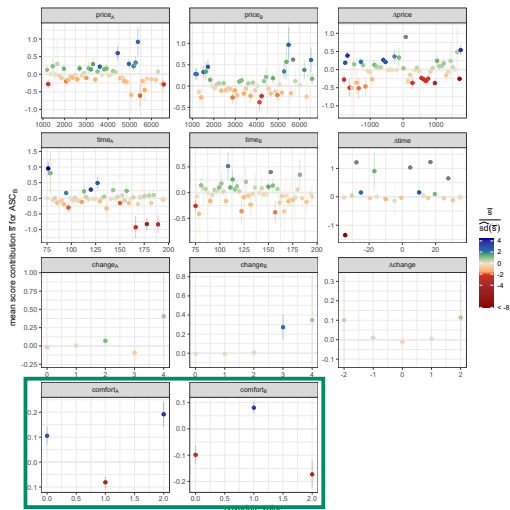
⁵Same utility formulation and mixed effects as used in Meijer and Rouwendal 2006

4. Application



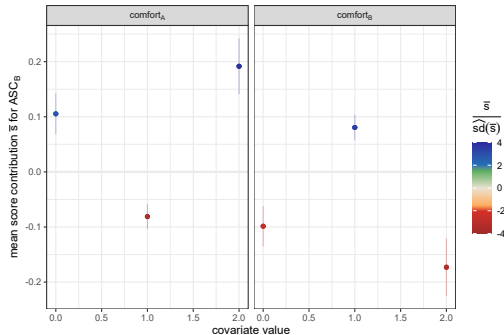
model	ll	AIC	BIC
initial model	-1 540.21	3 094.43	3 136.30
- mixed effects + ASC	-1 723.84	3 457.67	3 487.59

4. Application



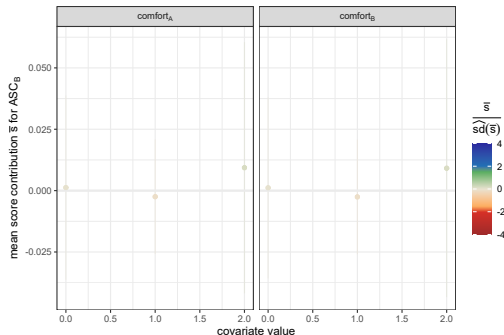
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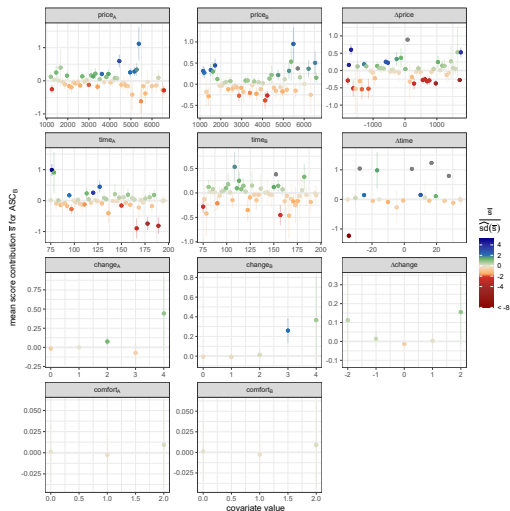
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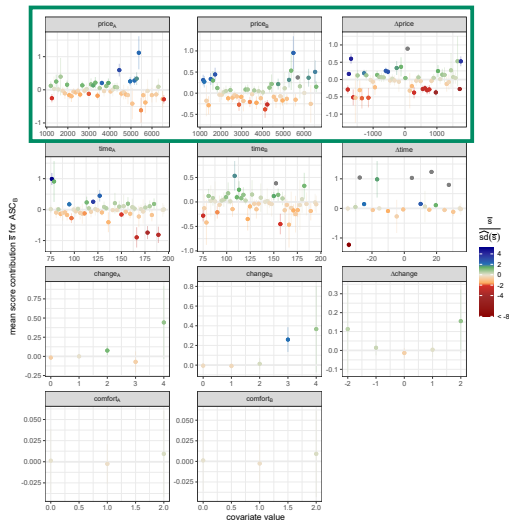
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+ as.factor(comfort)	-1 693.78	3 399.55	3 435.45

4. Application



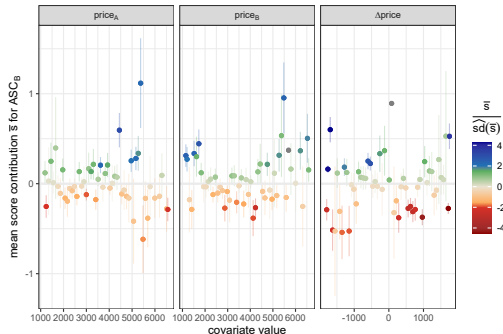
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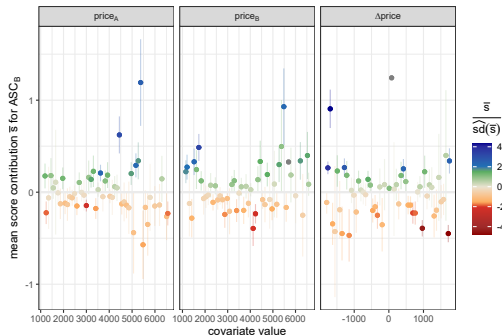
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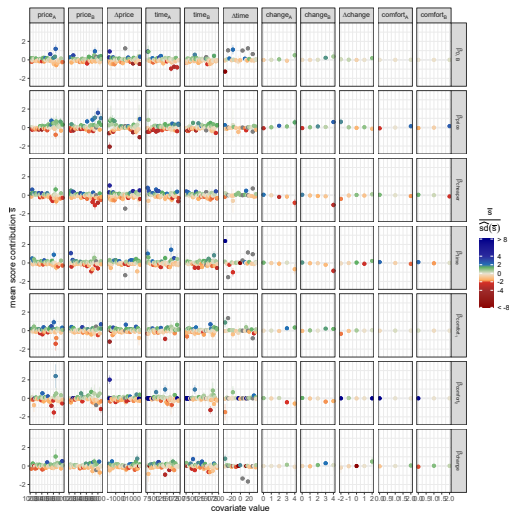
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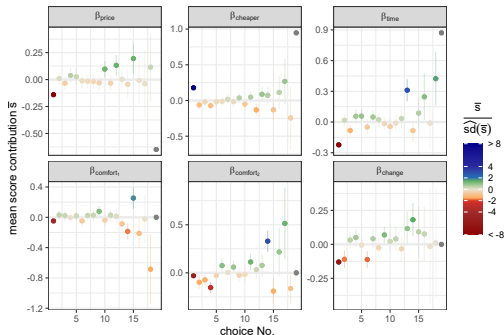
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+ $I(\Delta\text{price} > 0)$	-1 668.41	3 350.81	3 392.69

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initial model	-1 540.21	3 094.43	3 136.30
- mixed effects + ASC	-1 723.84	3 457.67	3 487.59
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+ as.factor(comfort)	-1 693.78	3 399.55	3 435.45
+ $I(\Delta price > 0)$	-1 668.41	3 350.81	3 392.69
+ mixed effects - ASC	-1 488.15	2 996.31	3 056.13
Δ	52.06	-98.12	-80.17

Mixed Models

Approaches for mixed models include

- use individual level gradients with mean covariate values
- investigate functional form of the utility function without mixing, add mixed effects for final model
- utilise **phantom gradients**: $\tilde{s}_{-n}^{(k)}(\hat{\beta}) = s^{(k)}(\hat{\beta}) - s_{-n}^{(k)}(\hat{\beta})$
 - $s_{-n}^{(k)}(\hat{\beta})$ is the gradient off all observations except of observation n
 - requires more computation time
 - not equivalent to gradient contributions
 - do not necessarily add up to zero

Outlook

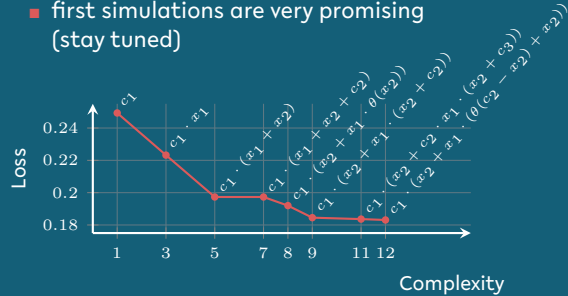
6. Outlook

R package:

- making visual gradient diagnostics easily accessible to practitioners
- extend to wider range of models (glm, mlogit, Rprobit, ...)
- extend to mixed models
- develop continuous estimation of expected conditional gradient contribution (nearest neighbour estimator)

Symbolic Regression:

- collaboration with **Dr. Kaja Balzereit**
- learn **functional form of the utility function** from **observational level gradients**
- first simulations are very promising (stay tuned)



$$U = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_2 + \beta_3 \cdot x_1 \cdot x_2 + \varepsilon$$

Bibliography I

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GitLab



App

Contact: sebastian.buescher@uni-bielefeld.de

Appendix

Proof of Lemma.

Using

- (a) that the sum of the choice probabilities over all alternatives has to equal one and thus the derivative of the sum is equal to zero ($\sum_{i=1}^J \partial_{\beta^{(k)}} P_{n,i}(\hat{\beta}) = 0$) and
- (b) that the expected value of the score is the sum of the conditional expectations of the score, conditional on the chosen alternative, times their choice probabilities,
- (c) the dominated convergence theorem ⁶, which, applied to sequence of random variables $\{X_n\}$ states that if (1) $\exists X$ such that $X_n \xrightarrow{P} X$ and (2) $\exists Y$ with $E(|Y|) < \infty$ such that $|X_n| \leq Y \forall n \in \mathbb{N}$, then $E(X_n) \rightarrow E(X)$, and
- (d) $\sum_{i=1}^J \frac{\partial_{\beta^{(k)}} P_{n,i}(\hat{\beta})}{P_{n,i}(\hat{\beta})} P_{n,i}(\beta_0) < \sum_{i=1}^J \frac{M}{\delta} P_{n,i}(\beta_0) = \frac{M}{\delta} < \infty$,

we get for a consistent estimator of $P_{n,i}(\beta_0)$:

Slides

⁶Çinlar 2011

Proof of Lemma (continuation).

$$\begin{aligned}
 \lim_{N \rightarrow \infty} E \left(s_n^{(k)}(\hat{\beta}; X_n, y_n) \mid X_n \right) &= \lim_{N \rightarrow \infty} E \left(\sum_{i=1}^J \frac{\partial_{\beta^{(k)}} P_{n,i}(\hat{\beta})}{P_{n,i}(\hat{\beta})} P_{n,i}(\beta_0) \mid X_n \right) \\
 &= E \left(\lim_{N \rightarrow \infty} \sum_{i=1}^J \partial_{\beta^{(k)}} P_{n,i}(\hat{\beta}) \underbrace{\frac{P_{n,i}(\beta_0)}{P_{n,i}(\hat{\beta})}}_{N \rightarrow \infty_1} \mid X_n \right) \\
 &= E \left(\lim_{N \rightarrow \infty} \underbrace{\sum_{i=1}^J \partial_{\beta^{(k)}} P_{n,i}(\hat{\beta})}_{=0} \mid X_n \right) \\
 &= 0.
 \end{aligned}$$



